

Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies
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PROFESSIONAL PROFILE

M.Sc. student in Quantum Technologies at TU Braunschweig with a B.Sc. in Computer Engineering (AI specialization). Hands-on work spans Python-based machine learning and computer vision, sensor/signal processing, MATLAB, and software automation. Academic depth covers quantum information/computing, quantum communication/photonics, semiconductor/device technologies, and information/coding theory, with projects in embedded systems and industry-university engineering.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering
Oct 2024 - Present
2 yrs 1 mo
Next: Internship → potential Thesis

Technische Universität Braunschweig — Braunschweig, Germany | 66/120 ECTS completed

Academic focus: Quantum Information & Computing • Quantum Communication & Photonics • Semiconductor & Quantum Devices • Information & Coding Theory

Selected academic work: QKD satellite communication • Nonlinear photonics analytical coursework • Coding-theory computational work • Semiconductor crystal defects

B.Sc. Computer Engineering (AI specialization)
Sep 2019 - Feb 2024
4 yrs 6 mos

Azad University of Mashhad — Mashhad, Iran

Academic focus: Software, Programming & Algorithms • AI, Machine Learning & Computer Vision • Embedded Systems & Hardware • Databases & Information Systems

EXPERIENCE

Independent Software & AI Projects — Münster, Germany
Oct 2025 - Present
1 yr

Software & AI Project Developer (Independent)

- Alongside my M.Sc. studies at TU Braunschweig, I have developed practical Python and AI applications addressing challenges in computer vision, sensor-data analysis, and machine learning. I continue this engineering work through the co-development of Career OS, an AI-assisted academic and career workflow platform, and plan to extend my projects into quantum computing and optimization.

Iran Khodro (IKCO) — Iran's largest automaker — Mashhad, Iran
Apr 2023 - Sep 2023
6 mos
Top 1

Computer Vision Developer — Bachelor's Thesis (Pouyesh Program)

- **Selection:** Ranked 1st among 50 top students from Azad and Ferdowsi Universities in Computer, Electrical and Mechanical Engineering, competing on different projects.
- **Industry challenge:** Developed a MATLAB vehicle detector for an assigned traffic-video problem.
- **Thesis & outcome:** The project became my Bachelor's thesis and led to an invitation to join IKCO's R&D team full-time; chose graduate studies instead.

Ferdowsi University of Mashhad — Mashhad, Iran
Mar 2023 - Aug 2023
6 mos

Project Contributor - Anodyne (University-Based Startup)

- **Project coordination & outreach:** Supported presentations, stakeholder communication and outreach for a dental-imaging initiative.
- **Industry events:** Assisted with on-site project presentation, attendee coordination, promotion and networking at dental and medical industry events.

Azad University of Mashhad — Mashhad, Iran

Teaching Assistant - Theory of Formal Languages and Automata
Sep 2023 - Jan 2024
5 mos

- **Teaching & assessment:** Led weekly exercise/problem-solving sessions for approximately 20 students and evaluated assignments and exams under course supervision.

Volunteer IT Support
Oct 2022 - Aug 2023
11 mos

- **On-site IT & camera systems:** Continued volunteering across university departments, troubleshooting PCs and installing, testing and connecting camera systems to local networks.

IT Support Intern
Jul 2022 - Sep 2022
3 mos

- **Hardware & network support:** Diagnosed/repaired desktops, replaced or upgraded components, configured Windows, and supported racks/cabling, LAN, and Wi-Fi.

TECHNICAL SKILLS

Programming	Python; C; C++; MATLAB
Machine Learning & Computer Vision	PyTorch; scikit-learn
Scientific Computing & Signal Processing	NumPy; SciPy; pandas; Matplotlib
Embedded Systems	ATmega32
Development Tools & Databases	Git; GitHub; GitLab; SQL

SELECTED TECHNICAL PROJECTS

AI SYSTEMS & WORKFLOW AUTOMATION — 2026

Career OS - AI-Assisted Academic & Career Workflow Platform (Link)

Google Apps Script; Google Drive API; Gemini/Groq; GitHub Actions

- Co-designed an evolving, privacy-first platform for postgraduate academic/career workflows; implemented persistent AI context, Drive ingestion, evidence provenance, queue/retry automation, AI API integrations and CI privacy checks.

MACHINE LEARNING & COMPUTER VISION — 2025

MRI Segmentation Inference Dashboard (Link)

PyTorch; Streamlit

- Integrated pretrained segmentation inference with preprocessing, weight loading, masks, overlays, and area visualization.

Gas Classification Pipeline and Simulation Dashboard (Link) — 2025

Python; scikit-learn; RandomForest; Streamlit

- Built statistical sensor-window features and a six-gas classification dashboard with class-probability output.

Activity Recognition ML Comparison (Link) — 2025

Python; PyTorch; scikit-learn; Poetry

- Compared RandomForest and 1D-CNN approaches for sensor activity recognition and configured lint/test quality checks.

Vehicle Detection & Traffic Video Analysis (Link) — 2023

MATLAB; Computer Vision; Image Processing

- Built separate MATLAB detection paths: HSV-based scene classification, Gaussian-mixture foreground segmentation and morphological filtering for daytime; binary thresholding and connected-component analysis for nighttime vehicle counting.

SIGNAL PROCESSING — 2023

Stereo Sound Direction (Link)

MATLAB; Signal Processing

- Analyzed stereo-audio window energy to identify the dominant left/right direction.

EMBEDDED SYSTEMS — 2021 - 2022

ATmega32 Digital Clock (Link) | ATmega32 Digital Safe (Link)

Embedded C; ATmega32; 7-segment display; matrix keypad; LCD

- Implemented coursework projects for a multiplexed digital clock and a keypad/password digital safe.

SELECTED MASTER'S TECHNICAL WORK

Coding Theory Computational Lab — 2026

MATLAB-based group work on cyclic-code structure, Hamming distance/correction radius, generator-matrix calculations, and recorded bit-error behavior versus SNR.

Information & Coding Theory — 2025 - 2026

Graded analytical work on block/prefix coding, Kraft sums, relative entropy, Markov processes, typical sets, and state-dependent channel concepts.

Nonlinear Photonics — 2025 - 2026

Analytical work on dispersion, phase matching, harmonic generation, nonlinear optical effects, fiber nonlinearities, and gain/threshold calculations.

Technical Presentations & Journal Club — 2025 - 2026

Prepared work on state-dependent channel coding, quantum communication networks, semiconductor crystal defects, and a plasmonic-metasurface literature review.

HONORS

B.Sc. Computer Engineering: Ranked 5th of 131 students	**2024 · Top 5**
Iranian University Entrance Exam (Konkoor, Mathematics): ~10,000 of ~144,500	**2018 · Top 7%**
Khorasan Razavi selective-school Entrance Exam: 12 of ~27,000	**2014 · Top 1%**

LANGUAGES & CERTIFICATES

Languages

German - A2 course completed; exam result pending; next: B1.1
English - TOEFL iBT 95/120
Persian - Native

Training (non-degree)

Sharif University of Technology - Quantum Programming Language (in progress)	2026
Ariaquanta Institute - Quantum Machine Learning	2025
Kaggle - Machine Learning; Introduction to Programming Using Python	2025
Azad University, Mashhad - Data & ML Bootcamp - 96 hours	2023
Jahad Daneshgahi, Khorasan Razavi - Python Programming - 40 hours	2021